

Geometry

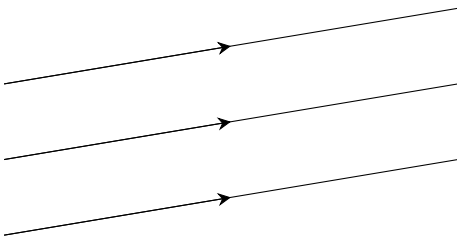
Covering

- Geometrical terms

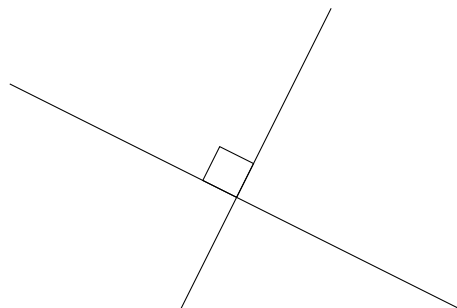
Geometrical terms

Name	Meaning
Point	A point is a location in space. It is a dot and usually named with a capital letter e.g A .
Line	A line is a straight path that extends in both directions infinitely. Its often named by the 2 points it connects, e.g Line AB , or its own name e.g Line L .
Vertex	A vertex is a corner point where two or more lines, rays, or edges/segments meet. e.g the corner of a triangle is a vertex.
Plane (O.E)	A plane is a flat, two-dimensional surface that extends infinitely in all directions. e.g a sheet of paper is a "plane", flat and 2d.
Parallel	A property for 2 or more lines which says they never meet or intersect, no matter how far. They are always the same distant apart. Its symbol is $\parallel$, where $AB \parallel CD$ means line AB and CD are parallel to each other.
Perpendicular	Two lines (or segments) are perpendicular if they intersect at a right angle (90°). Its symbol is $\perp$, where $AB \perp CD$ means line AB and CD are perpendicular to each other.
Perpendicular Bisector (O.E)	A perpendicular bisector is a line that cuts a line segment into 2 equal halves, and makes a 90° also.

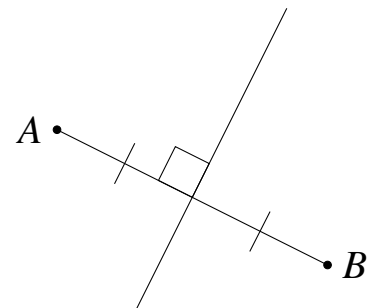
Parallel



Perpendicular



Perpendicular Bisector.



Patreon

Haris Rehman

Youtube

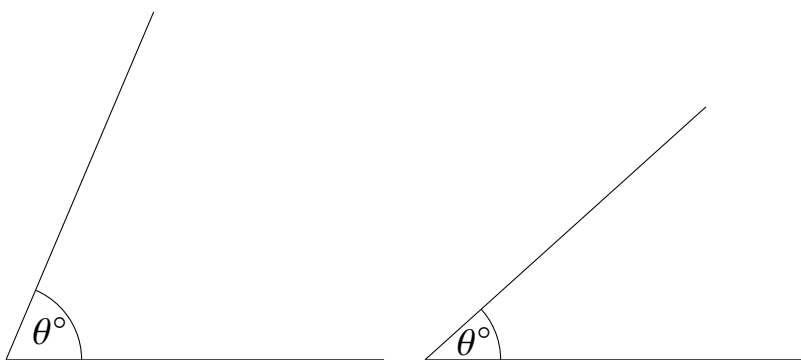
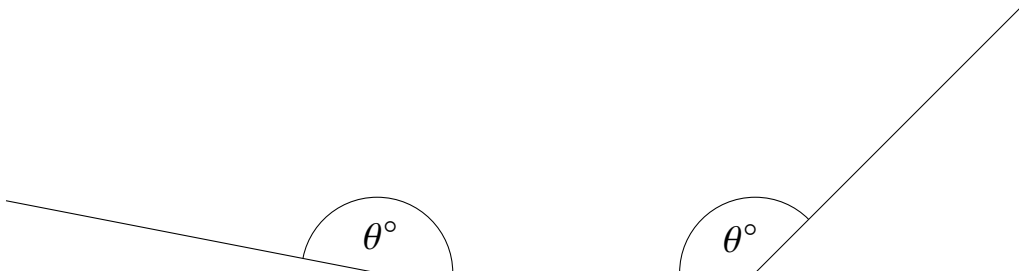


Angles of Polygons

Covering

- Angles

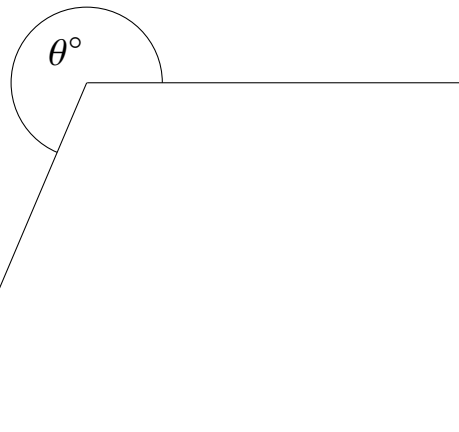
Types of Angles

Name	Meaning
Angle	The length of the arc between two lines which meet at a vertex, is the angle in degrees $^{\circ}$. The unknown variable commonly is θ .
Right Angle	When the angle, $\theta = 90^{\circ}$, that is a right angle.
Acute Angle	When the angle, $0^{\circ} < \theta < 90^{\circ}$, then θ is said to be an acute angle. Remember, cute = small things, i.e its <u>a cute</u> angle.
Obtuse angle	 When the angle, $90^{\circ} < \theta < 180^{\circ}$, then θ is said to be an obtuse angle. Remember, for "Ob" sound we open the mouth to an obtuse angle for it. 

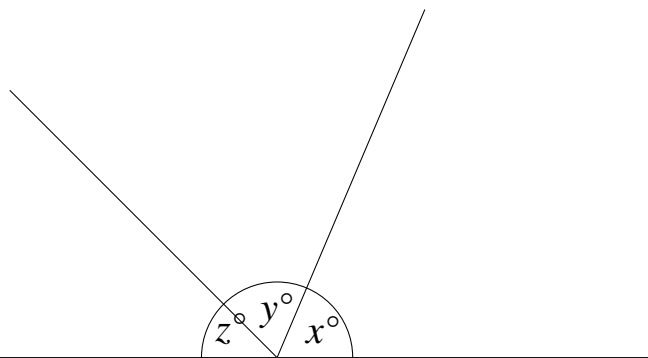


Reflex Angle

When the angle, $180^\circ < \theta < 360^\circ$, then θ is said to be reflex angle. Remember, reflex as in angle past your "D"/protractor.

**Angle Properties****Name****Meaning****Angle on a Line**

The angles on a straight line add up to 180° . e.g $x + y + z = 180^\circ$.

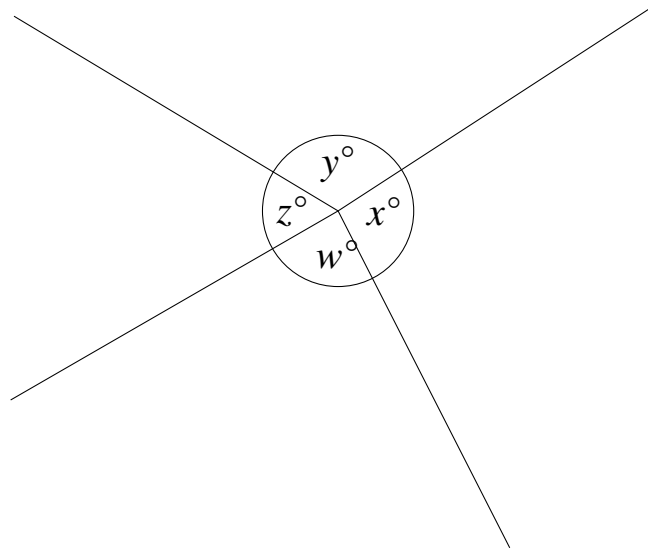
**Total Line Angle**

Acute + Obtuse = 180° . Here x is acute and $z + y$ is obtuse, $x + (y + z) = 180$



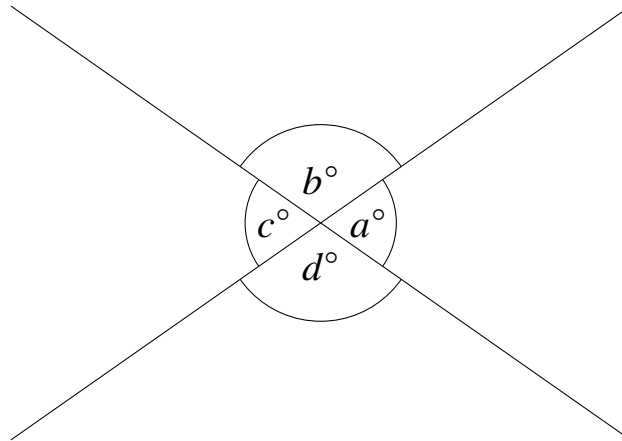
Angle at a Point

The angles at a point add up to 360° . e.g $x + y + z + w = 360^\circ$.



Vertically Opposite Angles

For two lines which intersect, they create the pair of angles; $a = c$, $b = d$. These pairs are written as "(Vert. Opp. $\angle$'s)".



Angles at parallel

When a line intersects 2 parallel lines, we get some angles with properties:

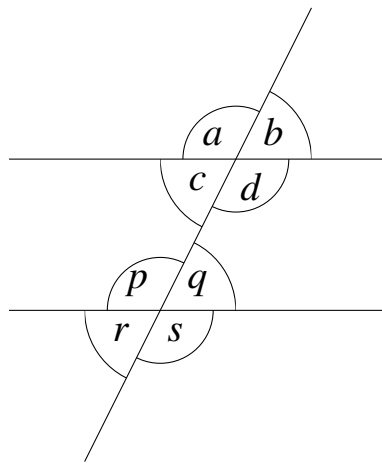
Vertically opposite angles are equal. $a = d$, $b = c$, $p = s$, $q = r$

Corresponding angles are equal. $a = p$, $b = q$, $c = r$, $d = s$

Alternate angles are equal. $c = q$, $d = p$

Sum of angles between parallel lines is 180° .

$c + p = 180^\circ$, $d + q = 180^\circ$

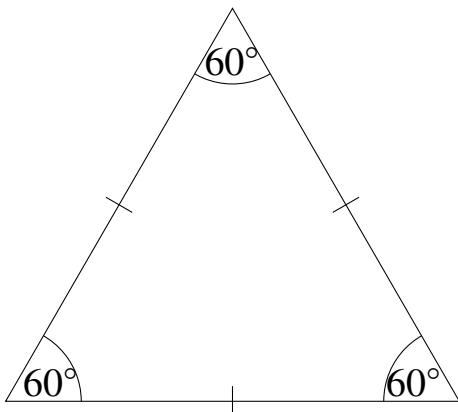
**Triangles**

There are 3 main types of triangles

Equilateral

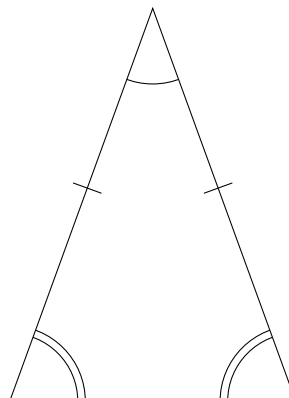
All sides are equal

All angles are 60°

**Isosceles**

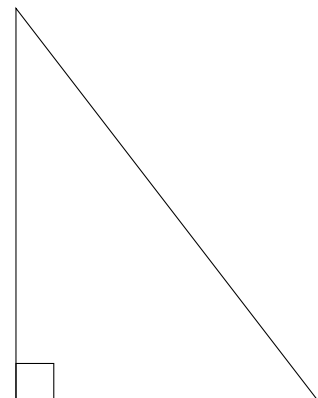
Two sides are equal

Base angles are equal

**Right Angle**

One angle is a right angle

Right angle = 90°



A scalene triangle is a triangle with different side lengths and angles.

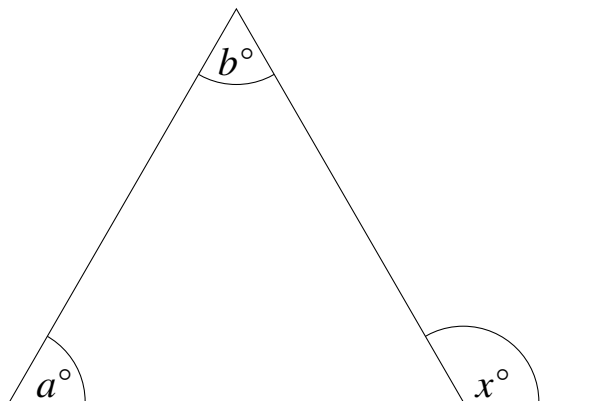


Angle properties of triangle

The sum of the angles in a triangle is 180° .

In every triangle, the greatest angle is opposite to the longest side. The smallest angle is opposite to the shortest side.

Exterior angle is equal to the sum of the opposite non-adjacent interior angles. i.e $x = a + b$



Polygons

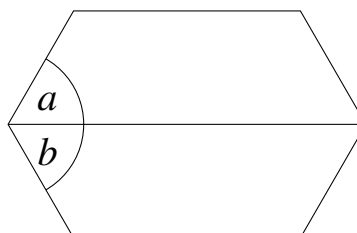
A closed figure with at least 3 sides is called a polygon.

Here are the names and side count of each polygon in the syllabus.

- Triangle has 3 sides
- Quadrilateral has 4 sides
- Pentagon has 5 sides
- Hexagon has 6 sides
- Octagon has 8 sides
- Decagon has 10 sides

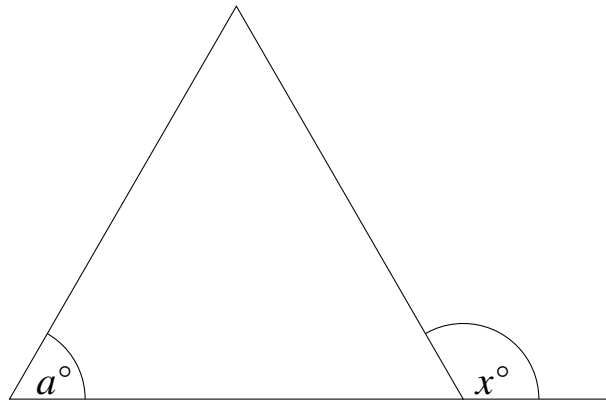
In a regular polygon, all sides and angles are same, else it is called a irregular polygon.

In a polygon, lines inside which a parallel to some side bisect the angle. e.g $\angle a = \angle b$



Interior and Exterior Angles

The interior and exterior angles in a polygon are, e.g. where a and x are. Interior angles are the angles made from the sides in a polygon. Exterior angles are made by extending one side, e.g. the base of this triangle, and measuring the angle made with the adjacent side.



The sum of interior angles are $180(n - 2) = \text{sum of interior } \angle\text{s}$, where n is the number of sides.

E.g. For a triangle, $n = 3$, so $180(3 - 2) = 180(1) = 180$.

To find a single interior angle, use $\frac{180(n - 2)}{n} = \text{single interior } \angle$, where n is the number of sides.

E.g. For a triangle, $n = 3$, so $\frac{180(3 - 2)}{3} = \frac{180}{3} = 60$.

This formula gives the average of all the angles in the polygon.

To find a single exterior angle, use $\frac{360}{n} = \text{single exterior } \angle$, where n is the number of sides.

E.g. for a triangle, $\frac{360}{3} = 120$.

If we have the exterior angle, but not the sides, we solve for n in the single exterior angle formula for $\frac{360}{\text{single exterior } \angle} = n$.

E.g. for single exterior $\angle = 15$, we see the number of sides are $\frac{360}{15} = 24$.

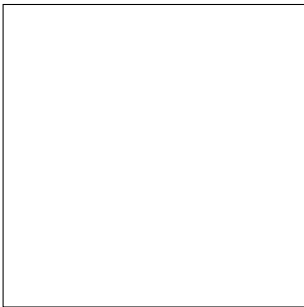


Quadrilaterals

Shapes which have 4 sides and an interior angle sum of 360 are Quadrilaterals. it include the shapes:

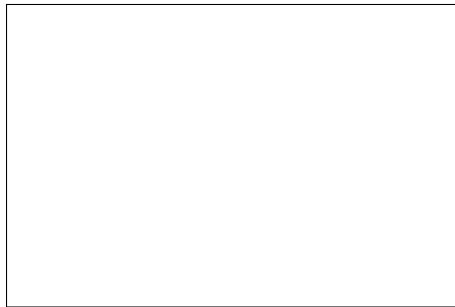
Square

All sides equal
All angles are 90°



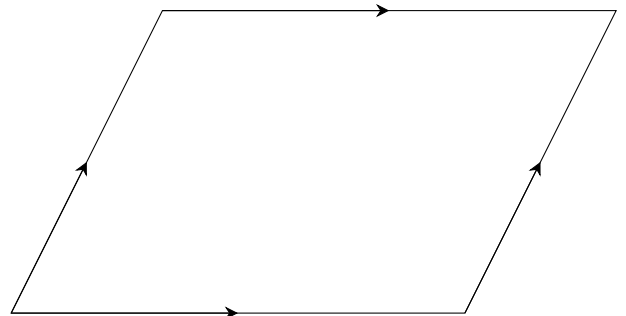
Rectangle

2 pairs of equal sides
All angles are 90°



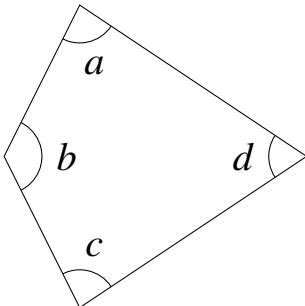
Parallelogram

2 pairs of parallel sides
Opposite angles are equal



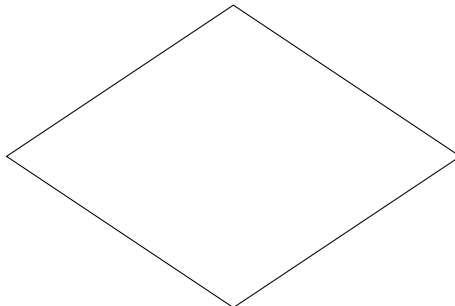
Kite

2 pairs of sides
of equal length
 $a = c, b \neq d$



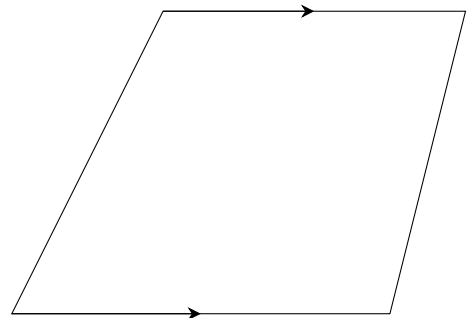
Rhombus

2 pairs of parallel sides
of equal length
Opposite angles are equal



Trapezium

Only 1 pair of parallel sides
Opposite angles sum to 180



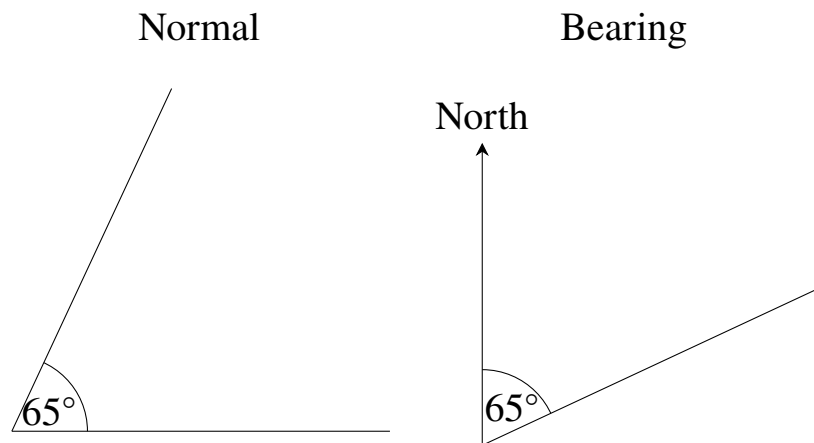
Bearing

Covering

- Scale drawings

Bearing

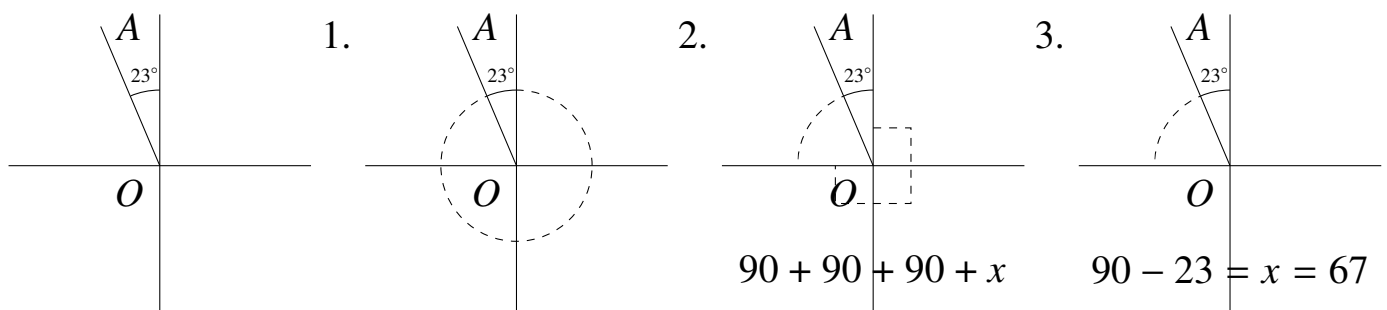
Bearing is just a method of measuring angles by the north direction line and rotating clockwise. E.g a normal 65° and a bearing of 065° are different, with the bearing being clockwise starting from the North arrow.



Bearings are always written in 3 digits, e.g 4° , 67° , $267^\circ \rightarrow 004^\circ$, 067° , 267°

The way they're asked is "bearing of A from B".

To measure, 1. first mark the angle, 2. check for any 90° , if so make the iconic "square" of 90° for them, 3. measure the remaining using a protractor or algebraically. E.g Find the bearing of A from O



Hence the bearing of A from O is: $90 + 90 + 90 + 67 = \boxed{337^\circ}$



Similarity and Congruence

Covering

- Similarity

Congruency

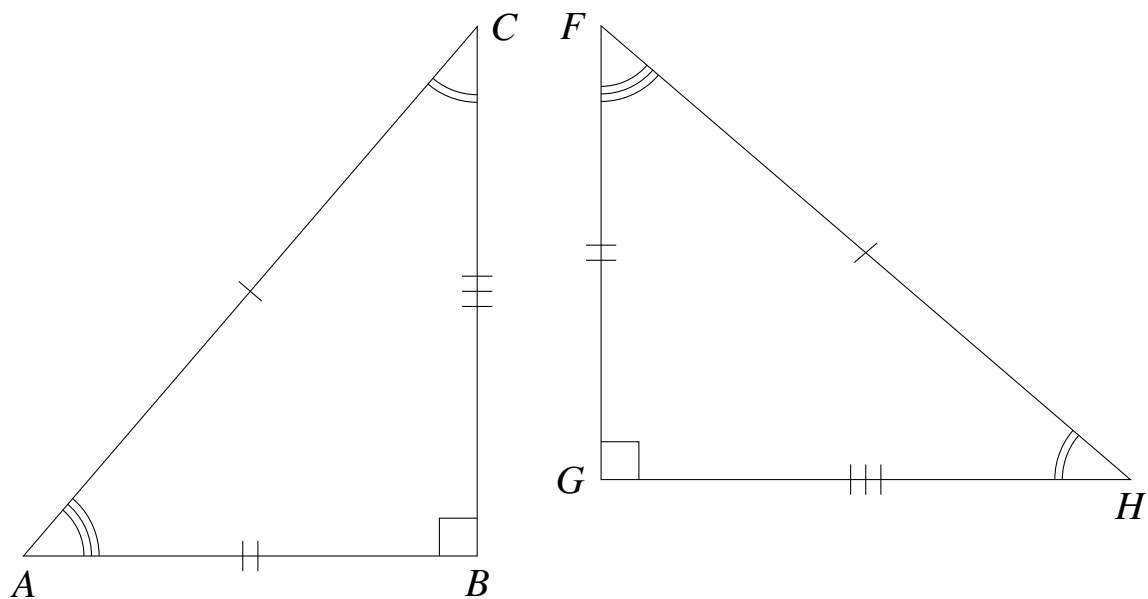
Congruency is to have equal measurement with the same order.

Two triangles are said to be congruent if they are same in every aspect.

For this order of sides and angles is compulsory. e.g $\triangle ABC \equiv \triangle FGH$,

here the side pairs we make are: $AB = FG$, $BC = GH$, $AC = FH$

here the angle pairs we make are: $\angle CAB = \angle HFG$, $\angle ABC = \angle FGH$, $\angle BCA = \angle GFH$



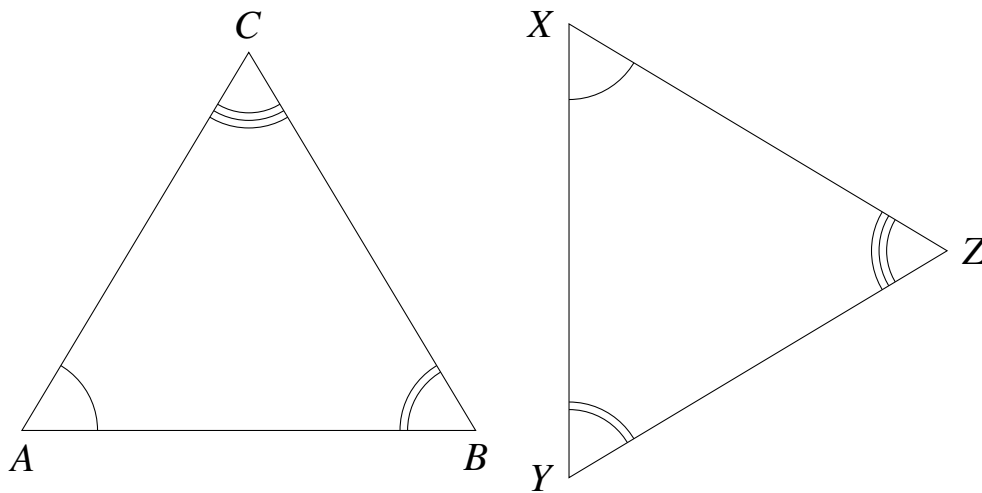
Similarity

Similarity is to have two figures increase or decrease by same proportion.

Two triangles are said to be similar if they have a pair of corresponding angles and ratios of the corresponding sides is the same. For this order of sides and angles is compulsory. e.g $\triangle ABC \equiv \triangle XYZ$,

here the side ratio pairs we make are: $\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ} = k$

here the angle pairs we make are: $\angle C\hat{A}B = \angle Z\hat{X}Y$, $\angle A\hat{B}C = \angle X\hat{Y}Z$, $\angle B\hat{C}A = \angle Y\hat{Z}X$



Proving Similarity

O levels and Extended

If you can show that one of the following conditions is true for any two triangles, then the two triangles are similar.

There are four cases to employ:

AAA

Angle Angle Angle is by showing the angles are equal using Angles Properties.

" $\triangle ABC$ is similar to $\triangle XYZ$ because $\angle \hat{A} = \angle \hat{X}$, $\angle \hat{B} = \angle \hat{Y}$ and $\angle \hat{C} = \angle \hat{Z}$ (AAA)"

SSS

Side Side Side is by showing the ratio of the side lengths is equal.

" $\triangle ABC$ is similar to $\triangle XYZ$ because $\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ} = k$ (SSS)"



SSAA

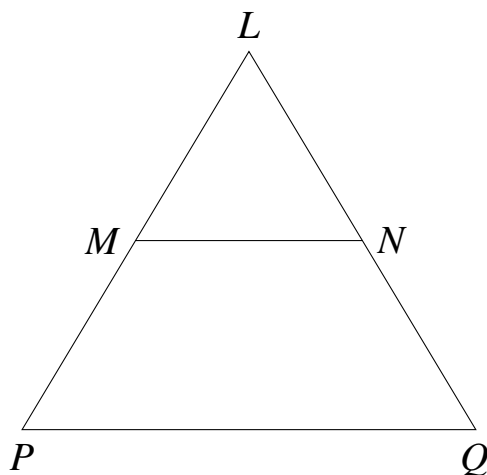
If you find, any 2 **Side** ratios being equal and 2 **Angle** pairs, then you say, e.g it could be.

" $\triangle ABC$ is similar to $\triangle XYZ$ because $\frac{AB}{XY} = \frac{BC}{YZ} = k$ and $\angle \hat{B} = \angle \hat{Y}$, $\angle \hat{C} = \angle \hat{Z}$ (SSAA)"

Area of similar figure**O levels and Extended**

We can use similarity to find Areas;

For $\triangle MLN$ is similar to $\triangle PLQ$

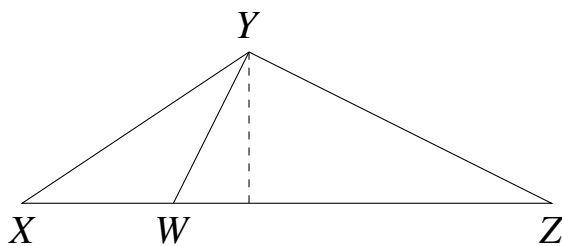


$$\frac{A_1}{A_2} = \left(\frac{l_1}{l_2}\right)^2$$

For l_1, l_2 being corresponding sides.

$$\begin{aligned} &\text{E.g} \\ &\frac{\text{Area of } \triangle MLN}{\text{Area of } \triangle PLQ} \\ &= \left(\frac{ML}{PL}\right)^2 = \left(\frac{LN}{LQ}\right)^2 = \left(\frac{MN}{PQ}\right)^2 \end{aligned}$$

For figures with 1 length matching, e.g their height or some other measure, we can use.



$$\frac{A_1}{A_2} = \frac{l_1}{l_2}$$

For l_1, l_2 being corresponding lengths.

$$\begin{aligned} &\text{E.g} \\ &\frac{\text{Area of } \triangle XYZ}{\text{Area of } \triangle WYZ} = \frac{XZ}{WZ} \end{aligned}$$

Volume of similar figure**O levels and Extended**

We can use similarity to find Volumes;

For this we use the formula $\frac{V_1}{V_2} = \left(\frac{l_1}{l_2}\right)^3$, for l_1, l_2 , being corresponding lengths.



Geometrical constructions

Covering

- Geometrical constructions

Construction

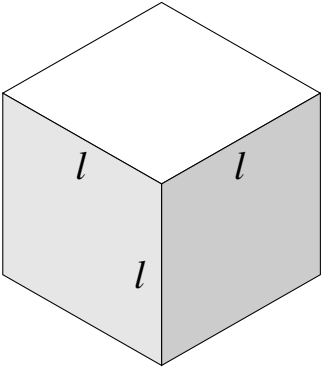
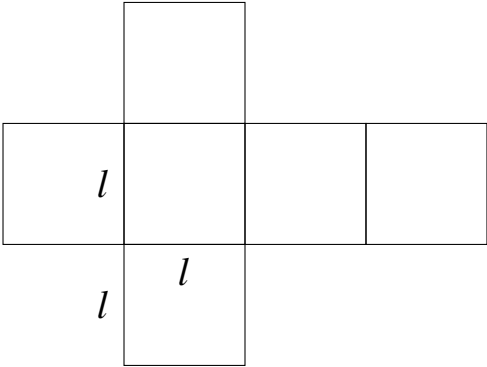
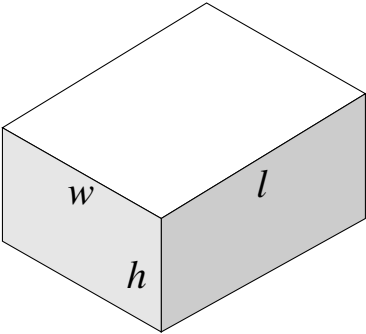
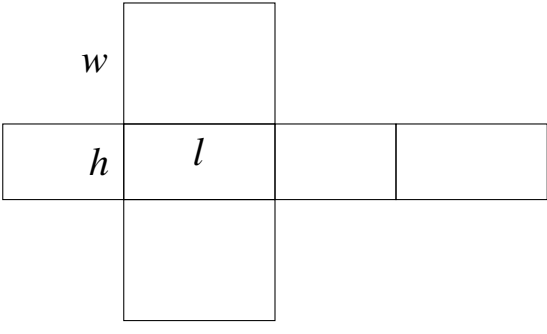
For this sadly text alone can not do justice and hence recommend *my* video

<https://youtu.be/WZ1zRq8FKh4?si=dXO3hMU6dZBBvC-g>

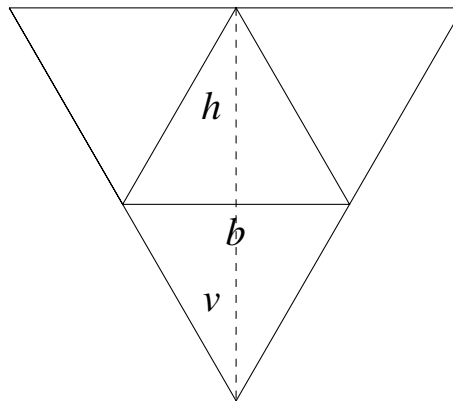
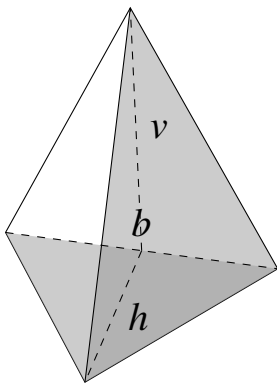
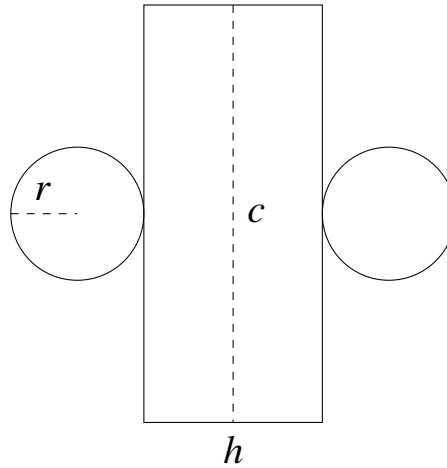
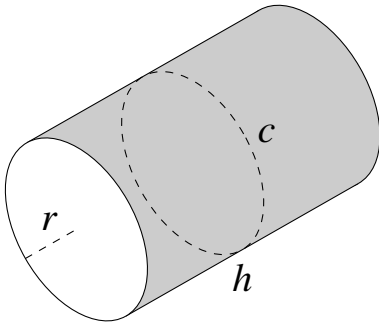
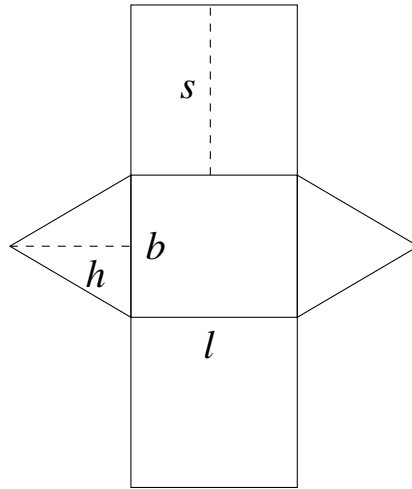
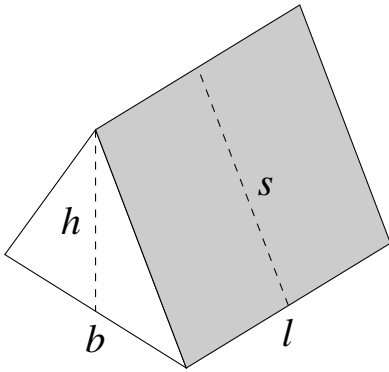
or search "Haris scale drawings" on youtube.

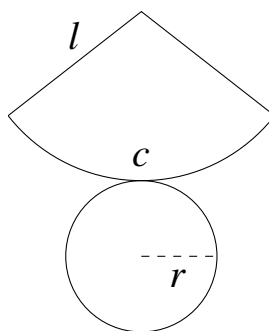
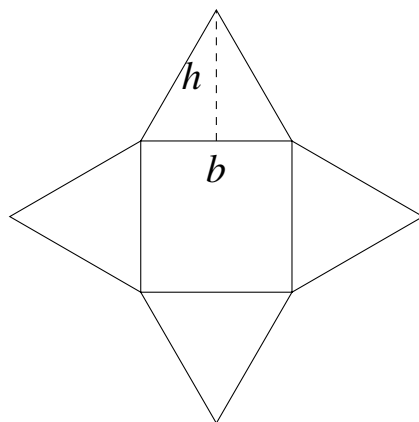
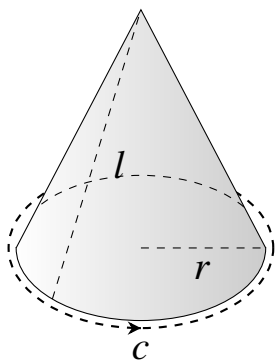
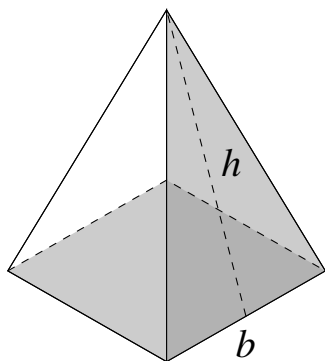
Nets of Solids

These solids, when you open them, give their nets. The variables show what length on the diagram is what on the net.

Folded	Net
	
	







Symmetry

Covering

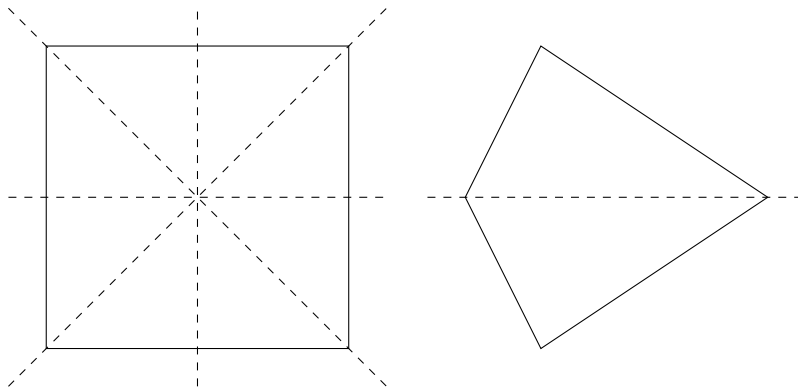
- Symmetry

Symmetry

Symmetry is about a shape being congruent, either it being even when mirrored by a symmetry line, or when rotated by the centre.

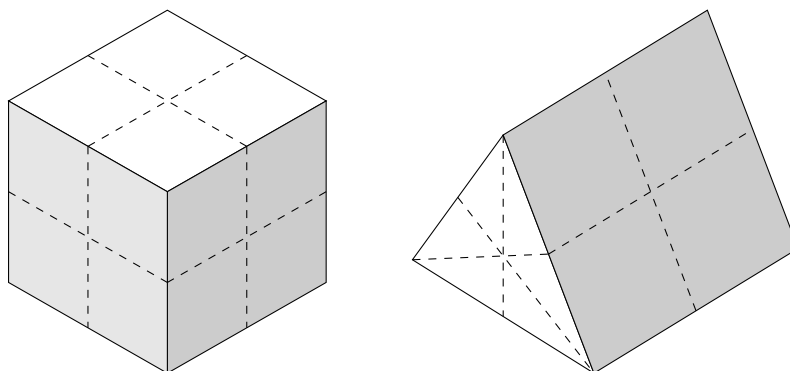
Line of Symmetry

A symmetry line divides a two-dimensional shape into two congruent shapes. E.g



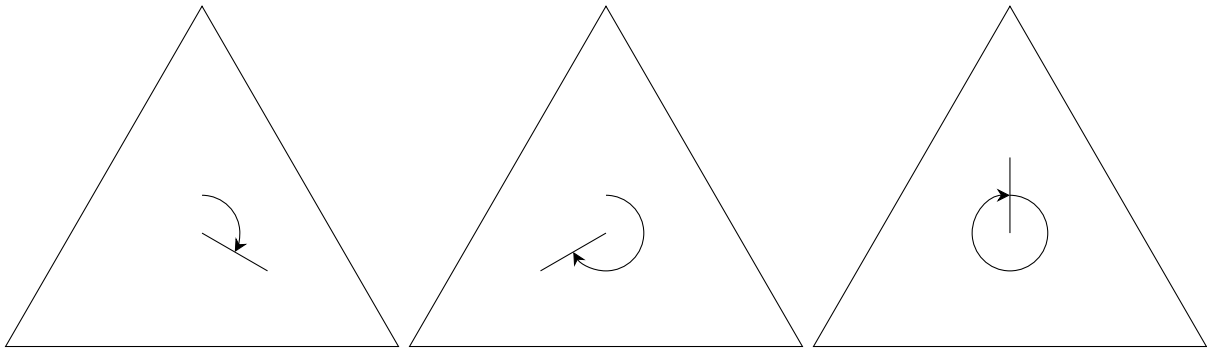
Plane of Symmetry

A symmetry plane divides a three-dimensional shape into two congruent solids. E.g
The dotted lines are the cuts by the symmetry planes.



Rotational Symmetry

A two-dimensional shape has rotational symmetry, if when rotated about the centre, and it fits the outline. The amount of times it fits the outline during a full revolution is called the order of symmetry. E.g a triangle has a rotational symmetry of order 3.



Tables and Numbers

Here are the Lines of Symmetry and Rotational Symmetry Order counts of the Quadrilaterals.

Shapes	Number of Lines of Symmetry	Order of Rotational Symmetry
Square	4	4
Rectangle	2	2
Parallelogram	0	2
Rhombus	2	2
Trapezium	0	1
Kite	1	1

These apply for any quadrilateral, special trapeziums can have at least 1 Line of Symmetry.

All regular polygons have same number of lines of symmetry, and order of rotational symmetry.

E.g A Octagon has 8 lines of symmetry and the order of rotational symmetry is 8.

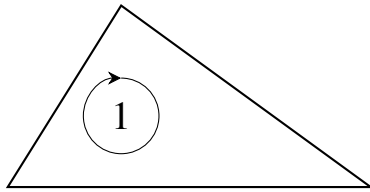
A Prism will have, Lines of Symmetry of cross section+1, Planes of Symmetry.

E.g a Pentagonal Prism (Pentagon as cross section) will have $(5+1)=6$ planes of symmetry.

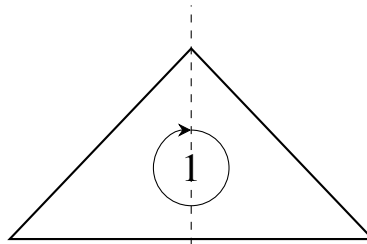


Here are all the shapes which can come. Dotted lines are the lines of symmetry, and the rotation is the order of rotational symmetry.

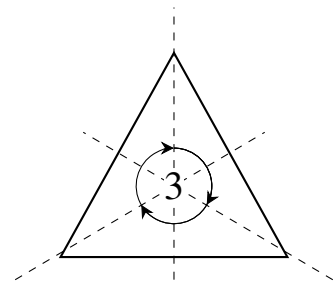
Triangles



Scalene

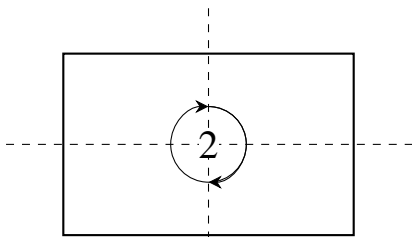


Isosceles

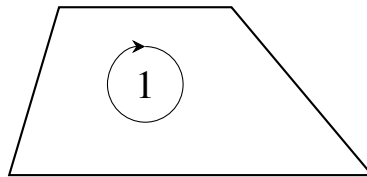


Equilateral

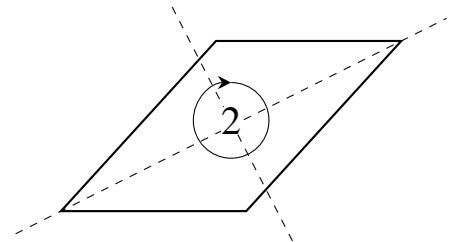
Quadrilaterals



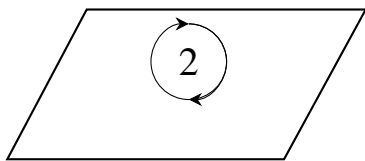
Rectangle



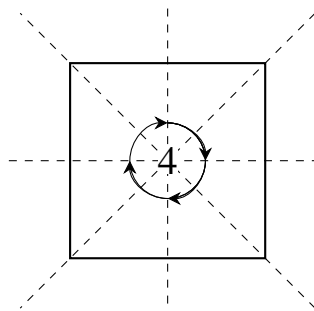
Trapezium



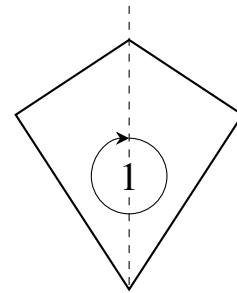
Rhombus



Parallelogram



Square



Kite

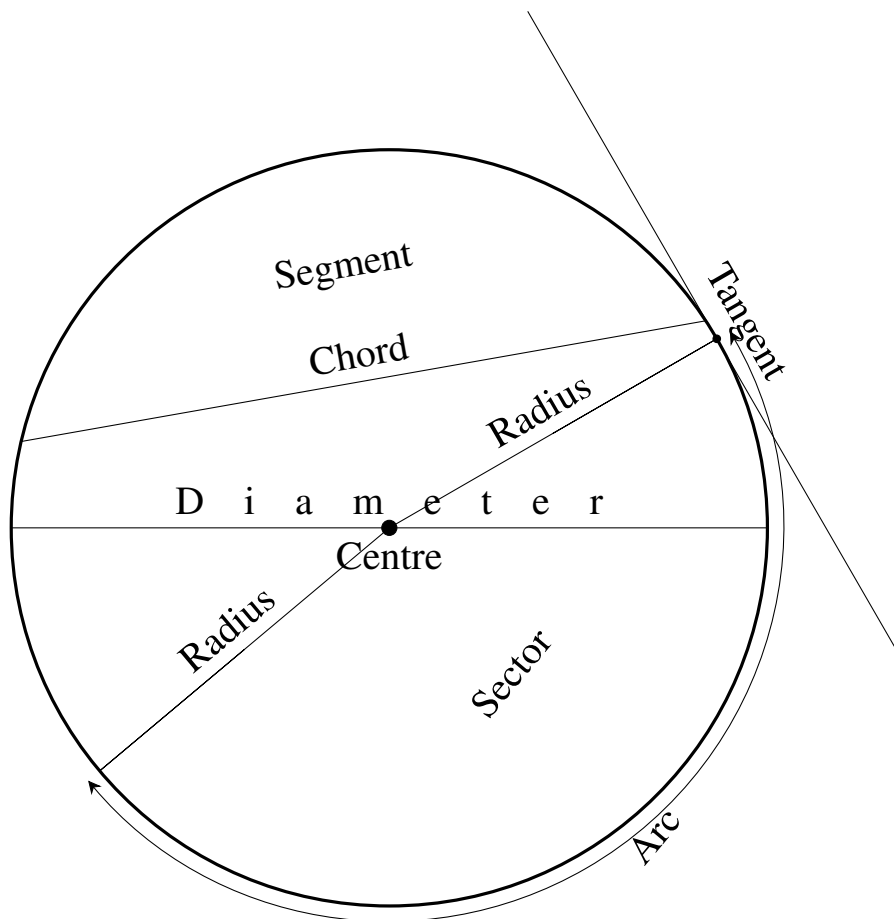


Circle Properties

Covering

- Circle theorems (I)
- Circle theorems (II)

Geometrical terms

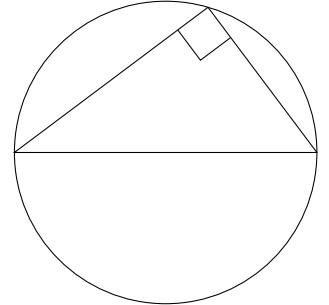
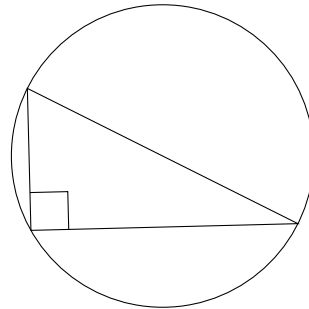
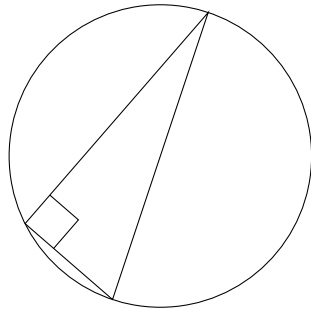


Circle Properties

Circle theorems (I)

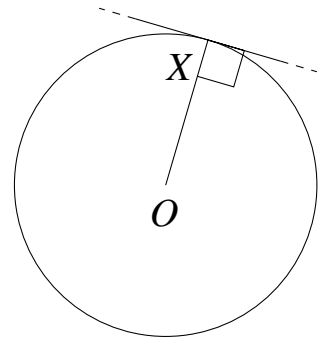
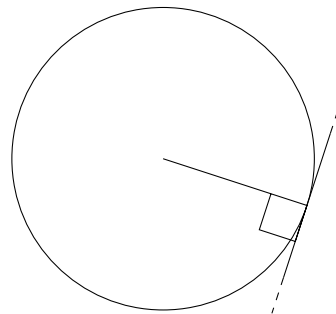
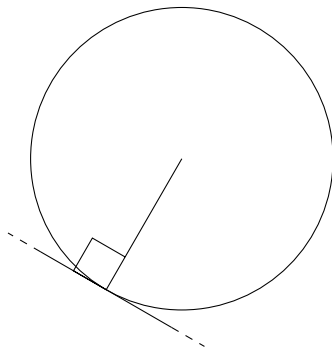
Angle subtended by diameter on circumference is 90° . For this you need to see and find a diameter, and then look for a triangle which has the longest side being the diameter of the circle.

E.g



Angle between tangent and radius is 90° . For this you need to look for a radius, e.g OX , where O is the origin and X is point on the circumference.

E.g

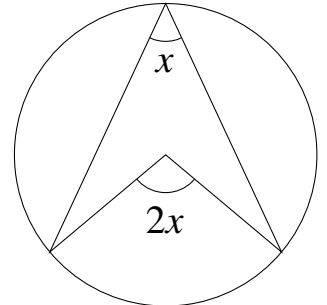
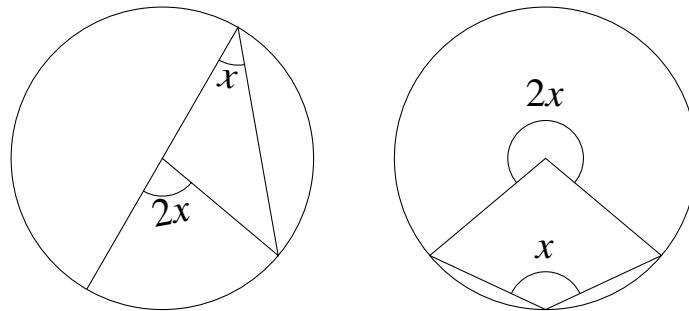


After this, the remaining rules are for

O Levels and Extended

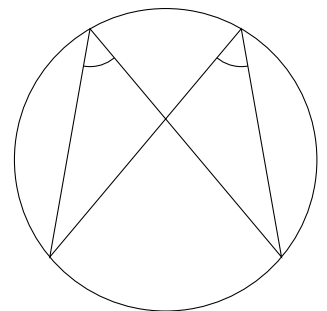
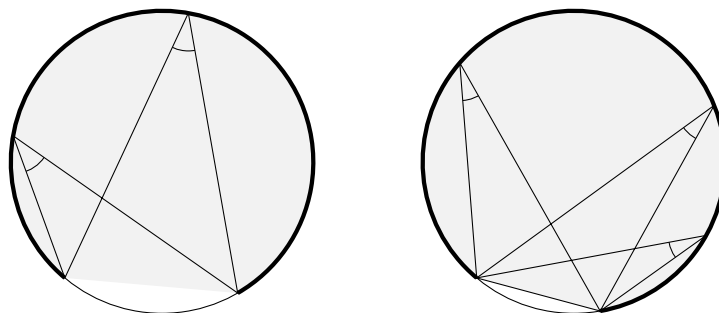
Angle at the centre is twice the angle at the circumference. For this you need to see if the points that help make the angle, also has line segments to the center of the circle.

E.g



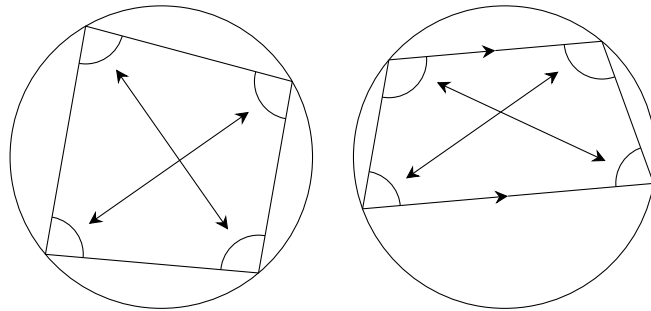
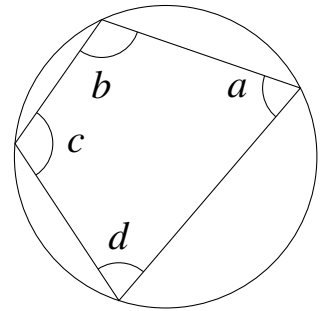
Angles in the same segment are equal. For this you need to see if the points that help make the angle, also has line segments to make another angle on the same arc/in the same segment. I have shaded the segment and thickened the arc line to see what we mean.

E.g



Opposite angles of a cyclic quadrilateral sum to 180° (supplementary). $a + c = 180^\circ$, $b + d = 180^\circ$. For this we need to look to the opposite side of the circle, and then detect a quadrilateral shape. Here I show arrows showing the opposing angles into few shapes.

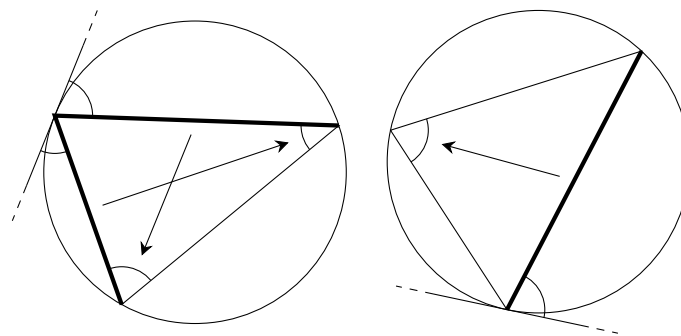
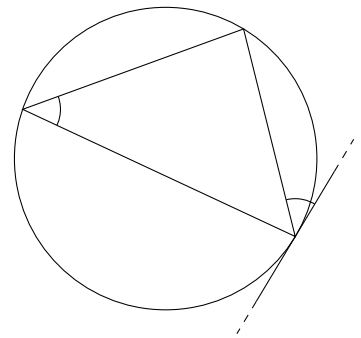
E.g



Alternate segment theorem

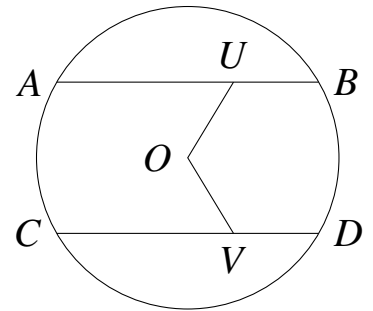
For a tangent at one point of a Δ , exterior angle with tangent is equal to non-adjacent opposite interior angle. For this 1. find the side of the angle the angle is made by with the tangent line also, that is the thick side of the triangles. 2. The angle opposite to that angle is the corresponding angle, that I drew a arrow to.

E.g

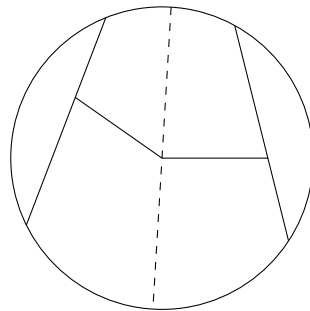


Circle theorems (II)

Equal chords are equidistant from the centre. If $OU = OV$, then $AB = CD$. This helps with symmetry questions stating 2 segments are same length, meaning there is a line of symmetry. Here it is shown dashed



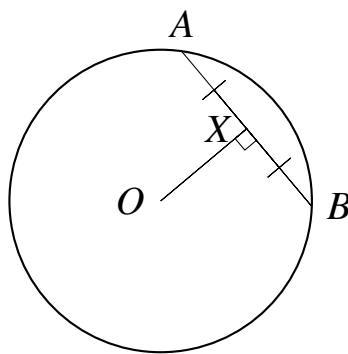
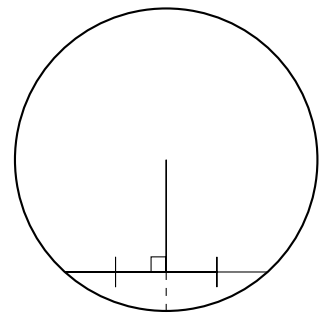
E.g



Questions can also use this property to setup working. E.g $AU : UB = 3 : 2$, $VD = 4\text{cm}$, find CV .

The perpendicular bisector of a chord passes through the centre. Questions can use this property to setup working.

E.g



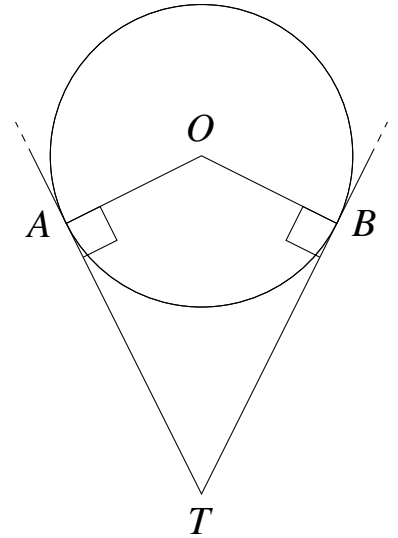
$OX = 3\text{cm}$ and $AB = 8\text{cm}$, find the radius of the circle.



Tangents from an external point are equal in length.

$AT = BT$, this property can be used for questions where you compute length, or to setup working.

$\angle O + \angle T = 180^\circ$, this property can be used to find either O or T , and to setup working.



E.g Given $AOB = 120$, $AB = 4cm$, find ATB .

AB and OT intersect to make X , $AXT = 90$

$$AX = 2cm$$

$$180 - 120 = 60, \quad \frac{60}{2} = 30 = AXT$$

$$AB \sin 30 = 2cm, \quad AB = 4cm$$

